

Solution
Mid-Semester Examination
Department of Applied Geophysics
Indian Institute of Technology (ISM), Dhanbad

Semester: Winter

Session: 2024-25

Time: 2 hours

Subject: Seismological Data Analysis

Subject Code: NGPD510 & GPD508

Marks: 30

Instruction, if any: All questions are compulsory. Provide to the point answer.

Sl. No.	Question	Marks			
1.	What is iterative time-domain deconvolution? Solution: Refer to class notes	2			
2.	What the following command would do? a. paste a.txt b.txt > c.txt b. join A.txt B.txt > C.txt c. awk '{ for (i = NF; i > 0; --i) print \$i }' d. awk 'NR == '\$n' {print \$0}' e. cut -d_ -f2 A.txt Solution: (i) To extract columns of information from different files and combine them into one file (ii) If two separate files share a common field they can combined with join (iii) Print fields in reverse order (iv) If NR is equal to shell variable 'n' print line Consider a file as follows (A.txt) that uses underscores to separate fields: <table><tr><td>Vs_7.2</td></tr><tr><td>Vp_11.3</td></tr><tr><td>Rho_6.6</td></tr></table> One could just extract the numeric values by: <pre>>> cut -d_ -f2 A.txt</pre>	Vs_7.2	Vp_11.3	Rho_6.6	1 x 5
Vs_7.2					
Vp_11.3					
Rho_6.6					
3.	What is shell? What are their various types? Solution: A shell is a command-line interface that allows users to interact with the operating system. It acts as an intermediary between the user and the operating system's kernel, enabling command execution, scripting, and automation. 1. Bourne Shell (sh) 2. Bash (Bourne Again Shell) 3. C Shell (csh) 4. Korn Shell (ksh) 5. Z Shell (zsh)	1 + 2			

4.	What is user defined variables? What is the significance of back quote, double quote, and single quote? Solution: Refer to class notes.	1 + 2
5.	a. What are the uses of following input command in vi editor: a, i, o, O Solution: In the Vi editor, the following input commands are used to enter Insert Mode, allowing users to insert or modify text: a (Append Mode) - Moves the cursor after the current character and switches to insert mode. - Example: If the cursor is on H in Hello, pressing a will allow typing after H. i (Insert Mode) - Moves the cursor before the current character and switches to insert mode. - Example: If the cursor is on H in Hello, pressing i will allow typing before H. o (Open a New Line Below) - Opens a new line below the current line and enters insert mode. - Example: If the cursor is on line 3, pressing o will create a new line at line 4. O (Open a New Line Above) - Opens a new line above the current line and enters insert mode. - Example: If the cursor is on line 3, pressing O will create a new line at line 2. b. What do these commands do in vi editor: (i) "a10yy (ii) yy (iii) "a9dd (iv) ., +10s/Banana/Mango Solution: (i) Yank 10 lines to buffer a (ii) Yank line to general buffer (iii) Delete 9 lines to buffer a (iv) replaces only the first occurrence of Banana with Kumquat in each of 11 lines starting with the current line (.) and continuing for the 10 that follow (+10). c. What does following command do on linux terminal (i) top (ii) ps -u username (iii) kill -9 PID (iv) echo \$SHELL Solution: □ top - Displays real-time system information, including running processes, CPU usage, memory usage, and more. - It helps monitor system performance and manage processes. - Press q to exit the top interface. □ ps -u username - Lists all the processes running under the specified username. - Example: ps -u john will display all processes running by the user "john".	2+4+4

	• Useful for monitoring user-specific processes. ❑ kill -9 PID • Forcefully terminates a process with the specified Process ID (PID). • The -9 signal (SIGKILL) immediately kills the process and does not allow it to save or clean up. • Example: kill -9 1234 will forcefully stop the process with PID 1234. ❑ echo \$SHELL • Displays the current shell being used. • Example output: /bin/bash (if using Bash) or /bin/zsh (if using Zsh). • Useful for checking which shell is active in the terminal.	
6.	Write a bash shell script to calculate factorial of a number using while loop. Solution: <pre>#!/bin/bash # Read number from user read -p "Enter a number: " num # Initialize factorial to 1 fact=1 n=\$num # Check if the number is non-negative if [\$num -lt 0]; then echo "Factorial is not defined for negative numbers." exit 1 fi # Calculate factorial using while loop while [\$n -gt 1] do fact=\$((fact * n)) n=\$((n - 1)) done # Print the result echo "Factorial of \$num is: \$fact"</pre>	2
7.	List various format for seismic data along with their brief details. Solution: Refer to class notes	3
8.	Using 'case' statement, write a bash script to identify various phases in a seismogram.	2

Solution:

```
#!/bin/bash -xv
input_phase=$1
case "$input_phase" in
    PKP)
        echo "PKP arrival"
        ;;

    SKS)
        echo "SKS arrival"
        ;;

    SPdKS)
        echo "SPdKS arrival"
        ;;

esac
```